

On the risk-management front, the Ivy 5 portfolio leverages a simple moving average rule. We explored this trading rule, as well as time series momentum (TMOM), in [Chapter 7](#). The simple moving average trading rule can be used across asset classes. [Figure 9.2](#) is a diagram of how the Ivy 5 with a moving average rule might look in practice.

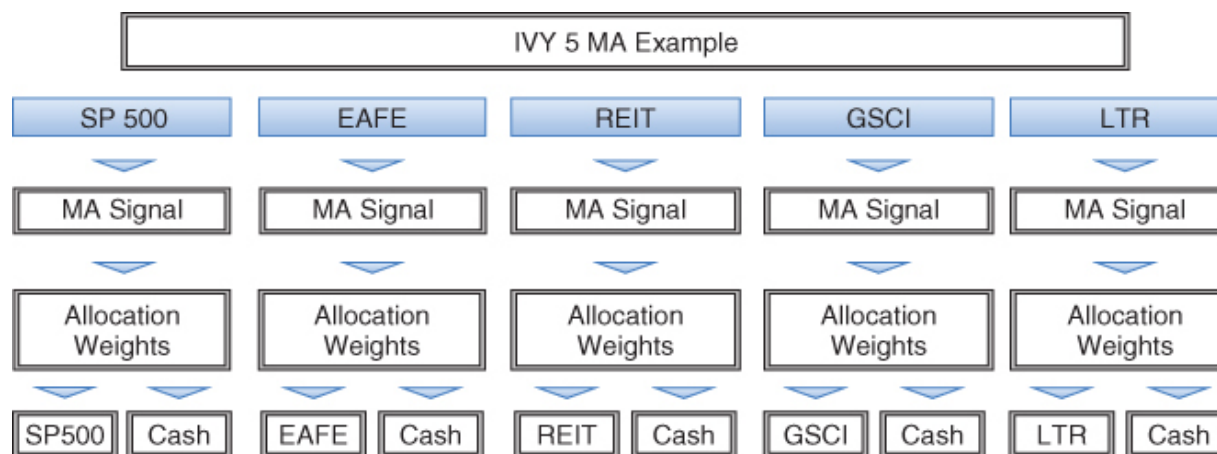


Figure 9.2 Ivy 5 with a Moving Average Rule

Ivy 5 with Moving Average Historical Performance

Our historical simulated time period is from January 1, 1976, to December 31, 2014, which is the longest period over which we can get data on the Ivy 5 asset classes. Results are gross of management fee and transaction costs. These are simulated performance results and do not reflect the returns an investor would actually achieve. All returns are total returns and include the reinvestment of distributions (e.g., dividends). Data are from Bloomberg and publicly available sources. Portfolios are annually rebalanced and the moving average rule is applied on a monthly basis. If an asset class is above its 12-month moving average at the end of the month, we maintain the position; otherwise, we sell the asset class and buy Treasury bills.

The following five asset classes are used in the backtest (referred to as the Ivy 5; see [Table 9.1](#)):

- SP500 = SP500 Total Return Index
- EAFE = MSCI EAFE Total Return Index
- REIT = FTSE NAREIT All Equity REITS Total Return Index